

Question

It is known that the chemical formula of a cluster compound **W** containing a p-block non-metallic element can be expressed as $A_2B_{14}Cl_{14}$, and its structure contains an electrically neutral fragment of B_{10} in the middle. The transition metal **A** is coordinated by 6 **B** atoms.

It is known that 5.346 g of the cluster compound **W** can be precisely obtained from 0.768 g of the transition metal element **A**, 2.688 g of the element **B**, and a certain amount of chloride **C** of **B** through a combination reaction at 250 °C. If **W** is heated to 280 °C in a hydrogen atmosphere, then **W** will decompose into the element **B**, 2.133 g of **D**, and 0.597 g of **E**, and 0.4032 L of gas **F** (341.1 °C, 1 atm). **D** is a coordination compound of **A**, belonging to the D_{3d} point group. After separating the element **B** from the above decomposition products, the remaining solid residue is further heated to a higher temperature to obtain 1.792 g of a ternary compound **G**, and two gases **C** and **F** are produced in a 1:1 ratio. **C** and **F** have the same elemental composition, and the volume of the mixed gas at 409 °C and 1 atm is 0.224 L. The molar mass of **E** is 0.28 times that of **D**. Then the following options are correct.

- A.** **W** decomposition yields chloride **C** of **B** containing 6 chlorines
- B.** The molecular weight of **F** is higher than that of **C**
- C.** **D** contains 6 chlorine atoms.
- D.** **G** can be represented as A_2B
- E.** 5.346 g **W** decomposes to produce 2.04 g of elemental **B**
- F.** The ratio of **G**, **C**, and **F** generated by heating after removing **B** is 2:1:1.
- G.** The ratio of **B** and **C** generated by the **W** decomposition is 2:1

H. All of the above options are incorrect.

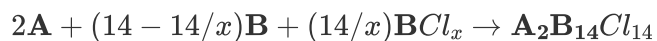
Answer

Correct Answer: **F**

Detailed Explanation

Solution process (Keep the original format, only change the uppercase letters to format):

The reaction of synthesizing **W** is a combination reaction. Let the chemical formula of **C** be BCl_x . We can try to write the equation of the reaction:



Obviously, the mass of **C** is $5.346 - 0.768 - 2.688 = 1.890(g)$

So we have:

$$\frac{2.688}{M(\text{B})} \cdot \frac{14}{x} = \frac{1.890}{M(\text{B}) + 35.45x} \times (14 - 14/x)$$

Simplifying, we get:

$$M(\text{B}) = \frac{35.45x}{0.703(x-1)-1}$$

CHECKPOINT

1 PTS

$$M(\text{B}) = \frac{35.45x}{0.703(x-1)-1}$$

Obviously $x > 2$. A simple discussion shows that when $x = 4$, $M(\text{B}) = 127.9$, corresponding to the *Te* element.

So the molecular formula of **C** is TeCl_4 .

CHECKPOINT

1 PTS

B is Te , **C** is $TeCl_4$

Therefore, option **A** is incorrect.

At the same time, it is obvious that

$$\frac{0.768}{2M(\mathbf{A})} = \frac{5.346}{2M(\mathbf{A}) + 35.45 \times 14 + 127.9 \times 4}$$

Solving for $M(\mathbf{A}) = 191.8$, corresponding to the Ir element

So the molecular formula of **W** is $Ir_2Te_{14}Cl_{14}$

CHECKPOINT

1 PTS

A is Ir , **W** is $Ir_2Te_{14}Cl_{14}$

According to the thermal decomposition experiment of the mixture **D** + **E**, the total mass of 1 : 1 **C** and **F** gases is $m = 2.133 + 0.597 - 1.792 = 0.938(g)$

The average molar mass of the mixed gas of **C** and **F** is $M = mRT/pV = 234.5 \text{ g mol}^{-1}$

It is known that **C** is $TeCl_4$, $M(\mathbf{C}) = 269.7 \text{ g mol}^{-1}$. Obviously, $M(\mathbf{F}) = 199.3 \text{ g mol}^{-1}$, and **F** is composed of the same elements as **C**. It is easy to know that **F** is $TeCl_2$

CHECKPOINT

1 PTS

F is $TeCl_2$

Therefore, option **B** is incorrect.

In the decomposition process of **W**, the mass of **F** produced is $m(\mathbf{F}) = pVM/RT = 1.597\text{ g}$

So $m(\mathbf{B}) = 5.346 - 2.133 - 0.597 - 1.597 = 1.019(g)$

CHECKPOINT

0.5 PTS

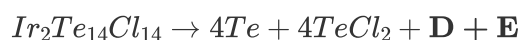
Mass of the simple substance produced during decomposition $m(\mathbf{B}) = 1.02\text{ g}$

Therefore, option **E** is incorrect.

Comparing, we can find that $n(\mathbf{B}) = n(\mathbf{F}) = 4n(\mathbf{W})$

Therefore, option **G** is incorrect.

Now try to write the decomposition equation of **W**. First, assume that 1 molecule of **D** and **E** are obtained:



D and **E** add up to $Ir_2Te_6Cl_3$. According to the question, $M(\mathbf{E}) = 0.28M(\mathbf{D})$

We can get $M(\mathbf{E}) = 298.6\text{ g mol}^{-1}$, corresponding to **E** being $IrCl_3$, so **D** is $IrTe_6Cl_3$, which also meets the symmetry of **D** in the question, that is, $[IrCl_3]$ is at the center of $[Te_6]$.

CHECKPOINT

1 PTS

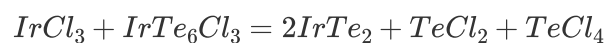
E is $IrCl_3$, **D** is $IrTe_6Cl_3$

Therefore, option **C** is incorrect.

The equation for the thermal decomposition experiment of the **D** + **E** mixture:



G is $IrTe_2$, and the reaction equation is:



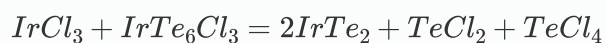
CHECKPOINT

1 PTS

The composition of **G** is $IrTe_2$

CHECKPOINT

1 PTS



Therefore, options *D*, *G*, and *H* are incorrect, and *F* is correct.